

ASME
IDETC-CIE 2026

INTERNATIONAL DESIGN ENGINEERING TECHNICAL CONFERENCES & COMPUTERS AND INFORMATION IN ENGINEERING CONFERENCE

HILTON AMERICAS HOUSTON | HOUSTON, TX, USA

ASME IDETC-CIE 2026
INTERNATIONAL DESIGN ENGINEERING TECHNICAL CONFERENCES & COMPUTERS AND INFORMATION IN ENGINEERING CONFERENCE

ASME® 2026 HACKATHON

Architecting the Future of Industrial Intelligence: AI for Integrated Design-to-Manufacturing

Virtual Kick-Off
August 13, 2026

Hilton Americas-Houston
Houston, TX, USA

In-Person Presentations and Awards Ceremony: August 23, 2026
At IDETC-CIE 2026

ASME IDETC-CIE 2026
INTERNATIONAL DESIGN ENGINEERING TECHNICAL CONFERENCES & COMPUTERS AND INFORMATION IN ENGINEERING CONFERENCE

ASME 2026 HACKATHON

Hilton Houston
Houston, TX

August 13 – 23, 2026
(Virtual and In-Person)

Register by August 11
Virtual Kick-off: August 16

Architecting the Future of Industrial Intelligence: AI for Integrated Design-to-Manufacturing

- **Problem 1:**

  Siemens Challenge: AI-powered CAM planning that turns CAD geometry into efficient, manufacturable CNC machining workflows.

  Problem 2:
  nTop Challenge: AI-driven design optimization for aircraft, balancing structural performance, aerodynamics, and mission requirements.

  Problem 3:
  AutoDesk Challenge: Visual language model systems that help engineers edit, refine, and improve existing 3D CAD models.



Participants compete for cash prizes!
https://event.asme.org/IDETC-CIE/Program/Student-Hackathon

Sponsorship:

SIEMENS
AUTODESK
nTop
ASME SETTING THE STANDARD

Thank you to our Sponsors!
ASME Computer & Information in Engineering Division (CIE), Siemens, nTop, and AutoDesk.

ASME Hackathon 2026 @ Houston/Online

https://event.asme.org/IDETC-CIE/Program/Student-Hackathon



ASME Hackathon 2026 @ Houston/Online

https://event.asme.org/IDETC-CIE/Program/Student-Hackathon

TAIBAH UNIVERSITY

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

NORTHWESTERN UNIVERSITY

UNIVERSITY OF CONNECTICUT 1881

MICHIGAN STATE UNIVERSITY

TARLETON STATE UNIVERSITY STEPHENVILLE, TEXAS 1899

UT DALLAS The University of Texas at Dallas

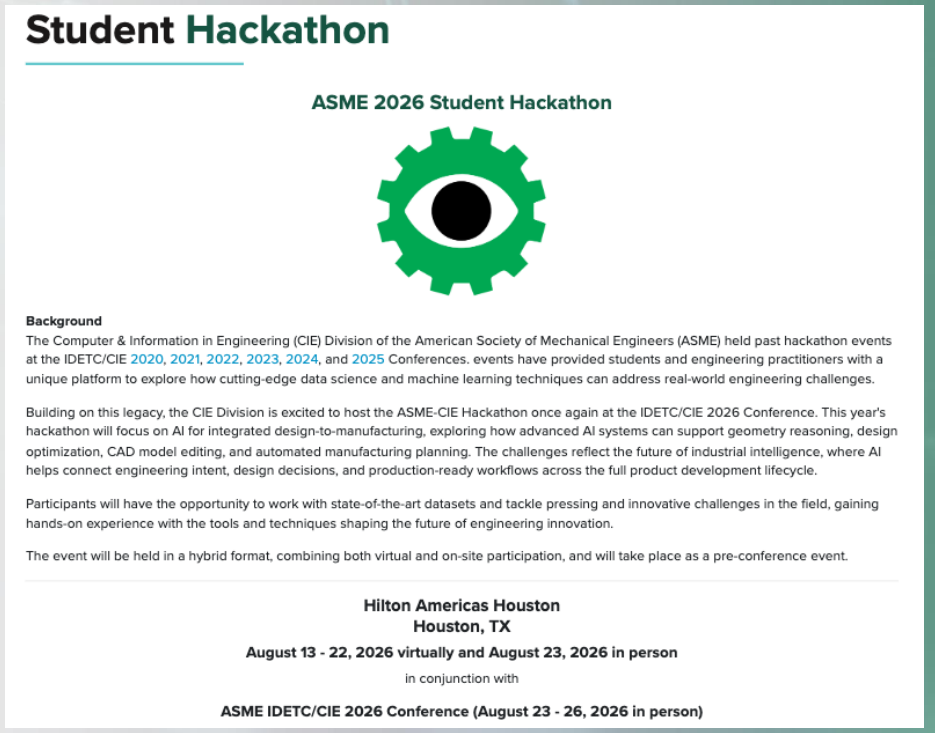
UNIVERSITY OF DELAWARE

NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY PAKISTAN

Massachusetts Institute of Technology

TECHNISCHE UNIVERSITÄT DARMSTADT

AUTODESK.



Student Hackathon

ASME 2026 Student Hackathon

Background

The Computer & Information in Engineering (CIE) Division of the American Society of Mechanical Engineers (ASME) held past hackathon events at the IDETC/CIE 2020, 2021, 2022, 2023, 2024, and 2025 Conferences. events have provided students and engineering practitioners with a unique platform to explore how cutting-edge data science and machine learning techniques can address real-world engineering challenges.

Building on this legacy, the CIE Division is excited to host the ASME-CIE Hackathon once again at the IDETC/CIE 2026 Conference. This year's hackathon will focus on AI for integrated design-to-manufacturing, exploring how advanced AI systems can support geometry reasoning, design optimization, CAD model editing, and automated manufacturing planning. The challenges reflect the future of industrial intelligence, where AI helps connect engineering intent, design decisions, and production-ready workflows across the full product development lifecycle.

Participants will have the opportunity to work with state-of-the-art datasets and tackle pressing and innovative challenges in the field, gaining hands-on experience with the tools and techniques shaping the future of engineering innovation.

The event will be held in a hybrid format, combining both virtual and on-site participation, and will take place as a pre-conference event.

Hilton Americas Houston
Houston, TX
August 13 - 22, 2026 virtually and August 23, 2026 in person
in conjunction with
ASME IDETC/CIE 2026 Conference (August 23 - 26, 2026 in person)

Hackathon Summary

- Three technical problems:
  - **Problem 1 - GeoCAiM: Geometry-driven CAM AI Planning** (Gaurav Ameta from SIEMENS)
  - **Problem 2 - AI-Driven Multidisciplinary Design Optimization** (Matthew Mueller and Ajay Prasad from nTop)
  - **Problem 3 - 3D CAD Model Editing with Visual Language Models** (Daniele Grandi from Autodesk)
- Awards for each problem:
  - 1st place: $1,400,
  - 2nd place: $700,
  - 3rd place: $350.
- **46+** participants; **25+** different institutions; **8+** different countries.

Agenda

First day (Thursday, 8/13/2026)	
Time (**EDT**)	Agenda
12:00 – 12:30	Hackathon Kick-off (Meeting link: **ASME 2026 Hackathon Kick-off**, Passcode: **498426**)
12:30 – 12:50	**Siemens:** Technical Problem #1 Presentation
12:50 – 13:10	**nTop:** Technical Problem #2 Presentation
13:10 – 13:30	**Autodesk:** Technical Problem #3 Presentation
13:30 – 14:00	Q&A
14:00 – 15:00	Participation socialization (Zoom & Slack) Zoom: **Meeting link**, Passcode: **498426**) Slack Channel: **Join Slack Workspace**
15:00 (8/13/2026) – 18:00 (8/15/2026)	Team formation & problem selection
Third day (Saturday, 8/15/2026)	
Time (**EDT**)	Agenda
20:00 (8/15/2026)	Deadline to fill out the team formation and problem selection form
Round 1 Submission day (Thursday, 8/20/2026)	
Time (**EDT**)	Agenda
9:00	Deadline for Round 1 Submissions (a 3-minute light presentation recording + other required deliverables)
10:00 (8/20/2026) – 11:00 (8/21/2026)	Round 1 Submissions Judging
Round 2 shortlist team announcement day (Friday, 8/21/2026)	
Time (**EDT**)	Agenda
12:00	Announcement of up to 5 teams advancing to Round 2 Live Finals

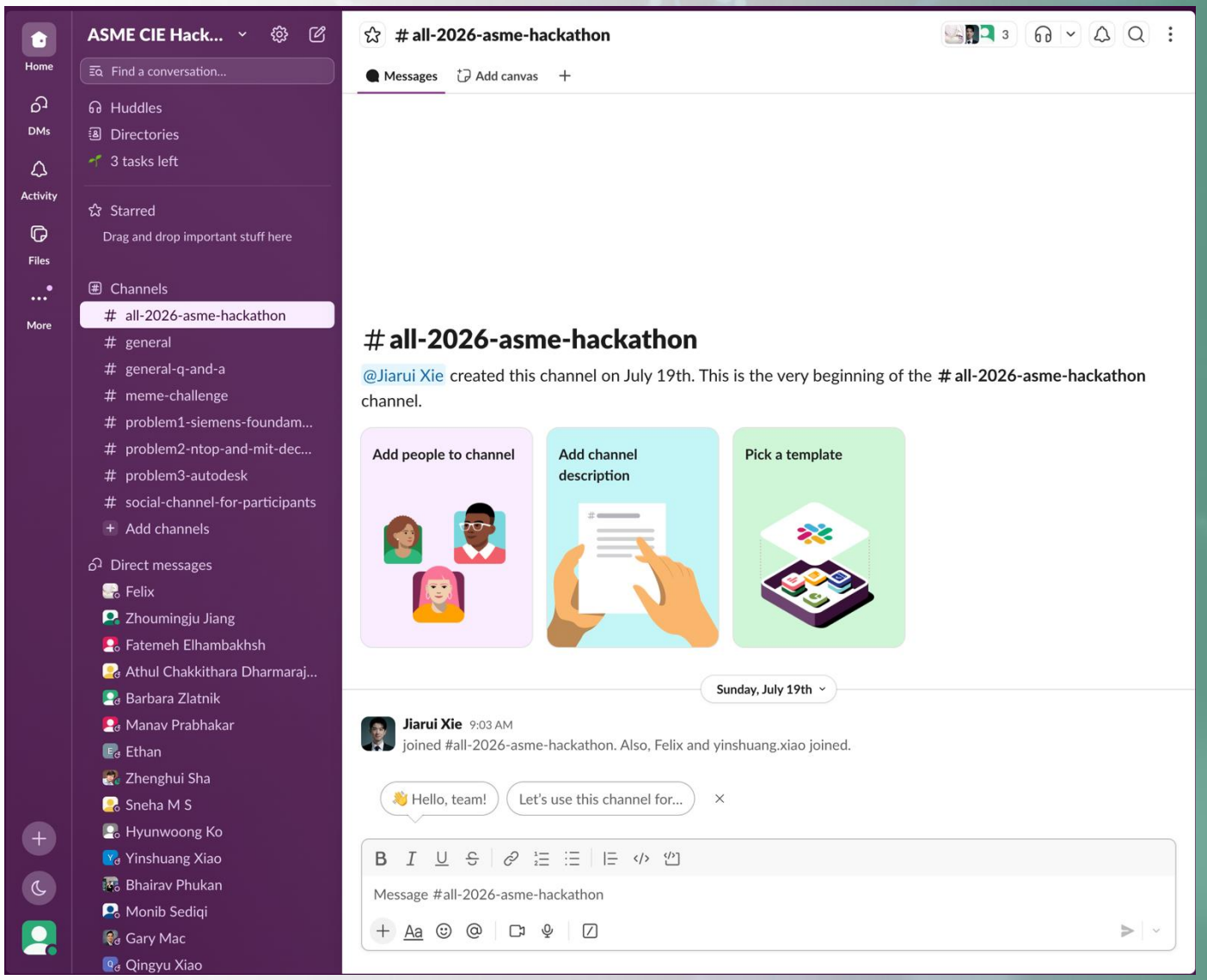
Previous team formation and problem selection form in your WelcomeEmail is unavailable, please use the link in this slide

Final day (Sunday, 8/23/2026)	
Time (**CDT**)	Agenda
7:00	Deadline for round 2 presentation slides submission
9:00 - 9:15	Check-in & working cycle (Meeting link: **ASME 2026 Final Day Check-in and Kick-off**, Passcode: **160139**)
9:15 - 9:30	Final day kick-off: Announce presentation schedule. Problems and judging criteria recap.
9:30 - 10:45	Presentation session
10:45 - 11:30	Judge discussion
11:30 - 13:00	Lunch
12:00 - 13:00	Awards & closing ceremony (Meeting link: **ASME 2026 Final Day Check-in and Kick-off**, Passcode: **160139**)

Communication Channels

Announcements will be made through **Slack Workspace** and **emails :**

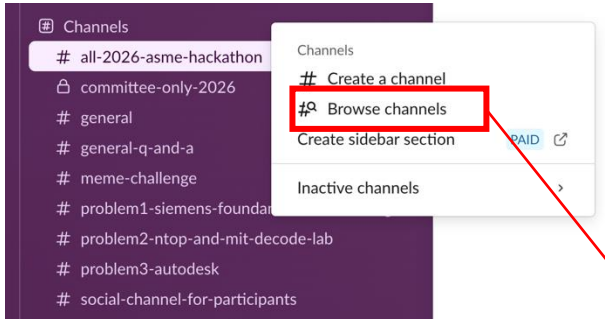
- Slack Channel: [Join Slack Workspace]
  - Major communication method
  - Different channels for different problems
  - Find teammates
  - MEME challenge
- Email: idetccie.seikm@gmail.com
  - You can also email your questions to the organizing committee.
  - Notifications might go to your spam.



The American Society of Mechanical Engineers®
ASME®

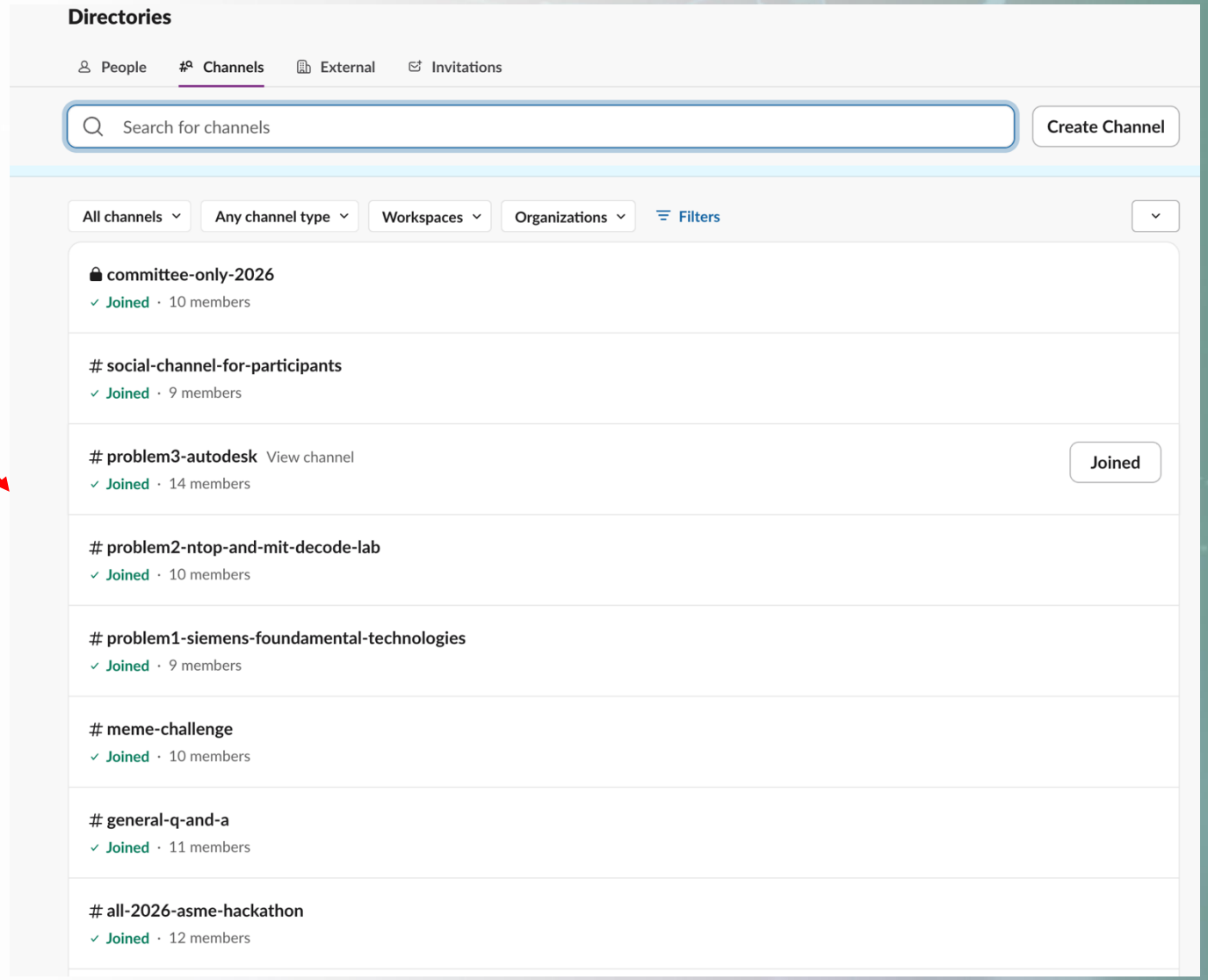
Communication Channels

- Right click “Channels”:



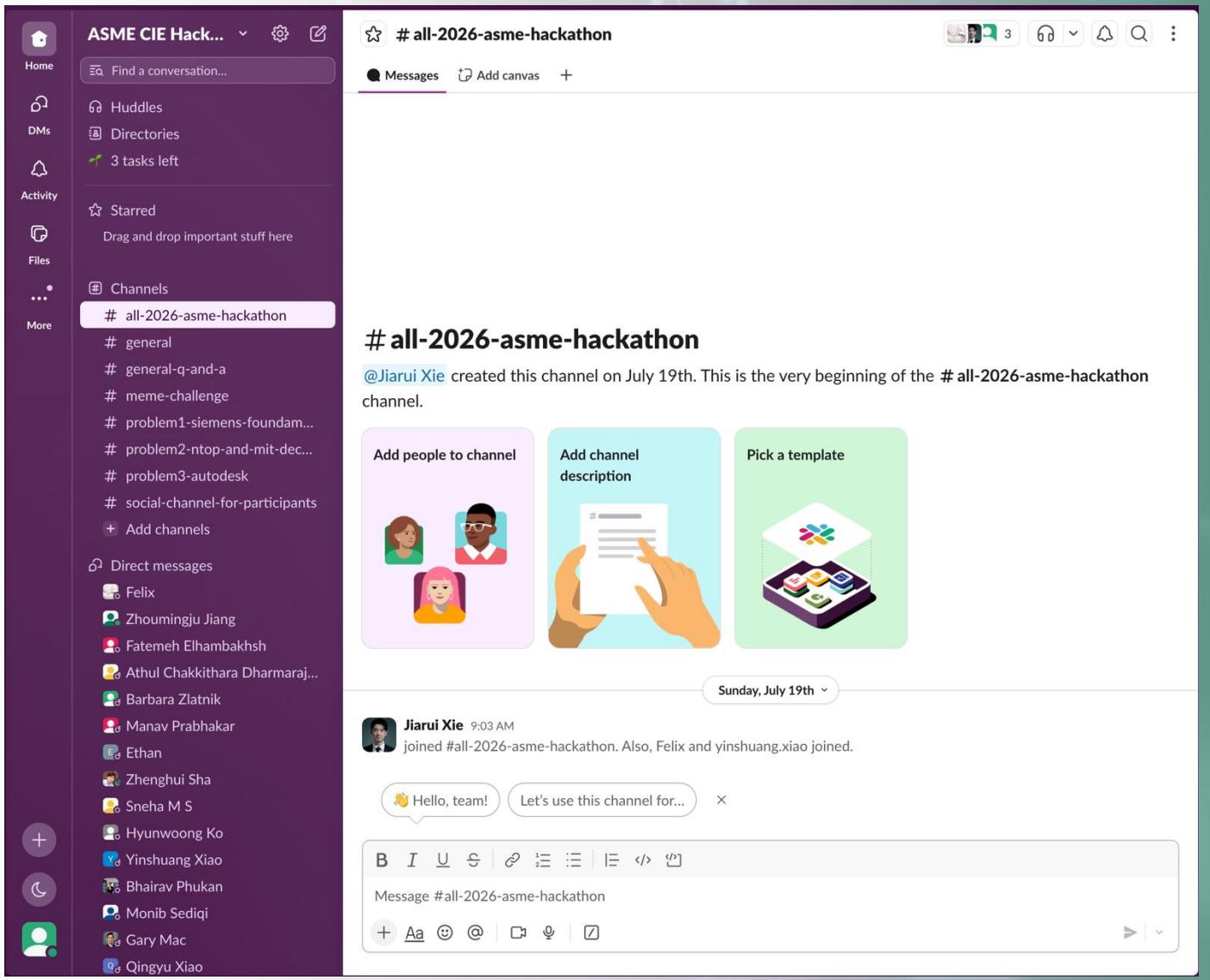
- Select “Browse channels”

- Please join all the channels listed in the right image



Teaming information

- Team size: **1 or 2**
- One team can work on **multiple problems**
- One person can only join one team
- The team numbers for each problem will be announced after your first-time selection
- You can switch to another problem or select multiple problems
- [Team registration & problem selection link]
- Deadline: **8 PM EDT (8/15/2026)**



Submission

- **Round One**
  - Step-1: Create a GitHub repository.
  - Step-2: Add the technical team member(s) to your GitHub repository.
  - Step-3: Prepare **3-minute** light presentation recording (Please embed Hackathon logo (pasted in slide 15) in your video)
  - Step-4: Upload video and final results on ***Thursday 9am (EDT), Aug 20, 2026***.
  - Rubrics depend on technical problems and judges.
  - Up to **5 teams** per problem will advance to Round 2 Live Finals (shortlist will be announced on ***Friday 12pm (EDT), Aug 21, 2026***)

Submission

- **Final Day (Round Two)**
  - Prepare and upload your final presentation slides (15 min: presentation + Q&A) by **7am (CDT), *August 23, 2026***
  - Final Presentation on ***Sunday, August 23, 2026***. Specific time/place and virtual meeting links to follow next week.
  - In-person: If you are at IDETC-CIE, there will be free lunch and coffee.
  - Virtual: Please attend virtual meetings to present at your scheduled time.

Financial Support Recipients

- Congratulations to the Travel Award & Fee Reimbursement recipients!
- The **fee reimbursement award** is conditioned upon completing the first-round submission and checking in with Prof. Yinshuang Xiao on Aug 23rd.
- The **travel award** is conditioned upon completing the first-round submission, participating final-day event in person, and checking in with Prof. Yinshuang Xiao on Aug 23rd.
- Financial support will be disbursed in the form of checks.

Travel Awards Recipients:

Yuewan	Sun
Ji	Qi
Qingqing	Mao
Hyunwoo	Kwon
Vaarij	Betala
Amir	Vahedi
Kiarash	Naghavi Khanghah

Fee Reimbursement Recipients:

Anwaruddin Siddiqui	Mohammed
Nirmal	Panta
Jipeng	Cui
Mithilesh	Kothandaraman
Hoang Anh	Nguyen
Samantha	Torres
Haoxuan	Mu
Lenin Goud	Athikam
Barbara	Sylvester
Sinan	Bank

ASME Hackathon 2026 MEME Challenge

An idea, behavior, or style that spreads by means of imitation from person to person within a culture and often carries symbolic meaning representing a particular phenomenon or theme. -Wiki







Using CV for Segmentation

Counting Pixels by Hand

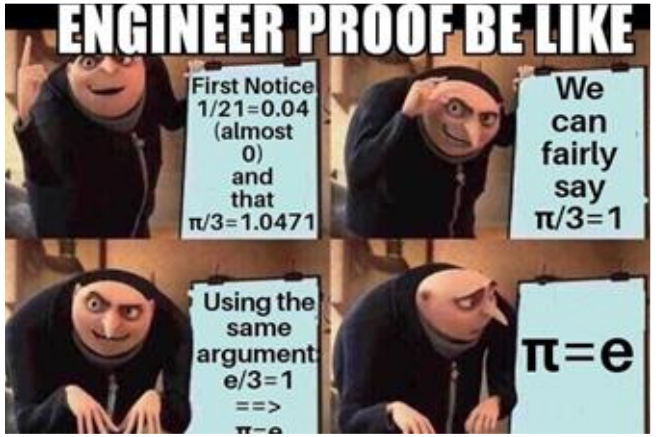


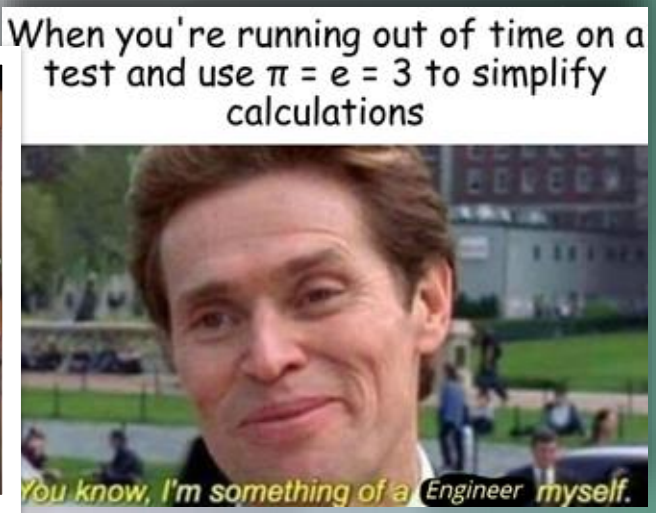


Gift card awards

- 1st place: $50
- 2nd place: $25
- 3rd place: $25

* Based on the number of reactions





Acknowledgment

- **Hackathon Participants**
- **Problem providers:**
  - SIEMENS (Dr. Gaurav Ameta, Athul Chakkithara Dharmarajan)
  - nTop (Dr. Matthew Mueller)
  - Autodesk (Daniele Grandi)
- **National Institute of Standards and Technologies (NIST)**
- **ASME Staff Support**: Barbara Zlatnik, Keli Bell-Cole.
- **CIE Division Executive Committee Representatives**: Gaurav Ameta, John Steuben
- **ASME Hackathon 2026 Committee Members**:
  - Prof. Yinshuang Xiao (Colorado State University), yinshuang.xiao@colostate.edu
  - Prof. Yaoyao Fiona Zhao (McGill University), yaoyao.zhao@mcgill.ca
  - Prof. Zhenghui Sha (University of Texas – Austin), zsha@austin.utxas.edu
  - Prof. Hyunwoong Ko (Arizona State University), hyunwoong.ko@asu.edu
  - Dr. Yan Lu (National Institute of Standards and Technologies), yan.lu@nist.gov
  - Dr. Jiarui Xie (McGill University), jiarui.xie@mail.mcgill.ca

Hackathon Logo

ASME® 2026
HACKATHON

Presentation Slides Template

https://event.asme.org/Events/media/library/resources/idetc-cie/IDETC-CIE-2026-Presentation-Template.pptx